

[Home](#)

[About Us](#)

[Studio Shodwe](#)

[Content](#)

[Contact Us](#)



CYBER SECURITY

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About Us

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Content

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CYBER SECURITY



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Content

- Cybersecurity is essential for protecting modern web applications from threats, unauthorized access, and data breaches.
- This report presents a passive vulnerability assessment conducted on the OWASP Juice Shop.
- The project provided hands-on experience in vulnerability scanning, risk analysis, and professional cybersecurity reporting.

INTRODUCTION



Definition of Cyber Security

Cyber Security is the practice of protecting computers, networks, devices, and data from digital attacks, theft, and unauthorized access.

It ensures that information stays confidential, accurate, and available — the three pillars known as the CIA Triad.

In today's world, cyber security is essential for safeguarding personal data, businesses, and critical infrastructure.

WHAT IS CYBER SECURITY?



Content

- Identify common web application vulnerabilities
- Perform passive vulnerability scanning
- Analyze exposed services and security weaknesses
- Classify vulnerabilities according to risk levels
- Recommend mitigation and remediation strategies
- Gain practical cybersecurity assessment experience

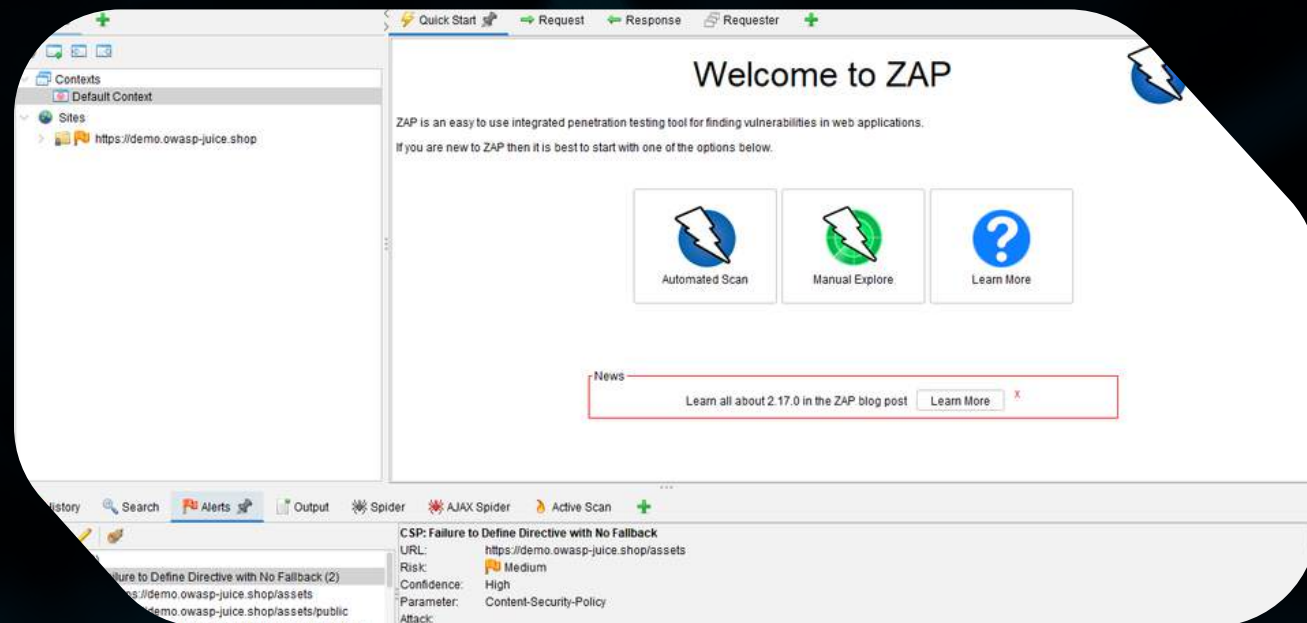
PROJECT OBJECTIVES



TOOLS USED

Cybersecurity Tools Used:

- Nmap — Network scanning and service discovery
- OWASP ZAP — Passive vulnerability scanning
- Browser Developer Tools — Web application inspection
- Windows Command Prompt — Scan execution
- Canva — Professional report and presentation design



Purpose: These tools were used to identify vulnerabilities, inspect web traffic, analyze security weaknesses, and document findings professionally.

TARGET INFORMATION



Content:

- Target Website: OWASP Juice Shop
- Target Type: Intentionally vulnerable web application
- Assessment Type: Passive vulnerability assessment
- Testing Environment: Authorized educational security testing
- Scope: Focused on publicly exposed vulnerabilities and misconfigurations (non-destructive testing)

WHY IT MATTERS



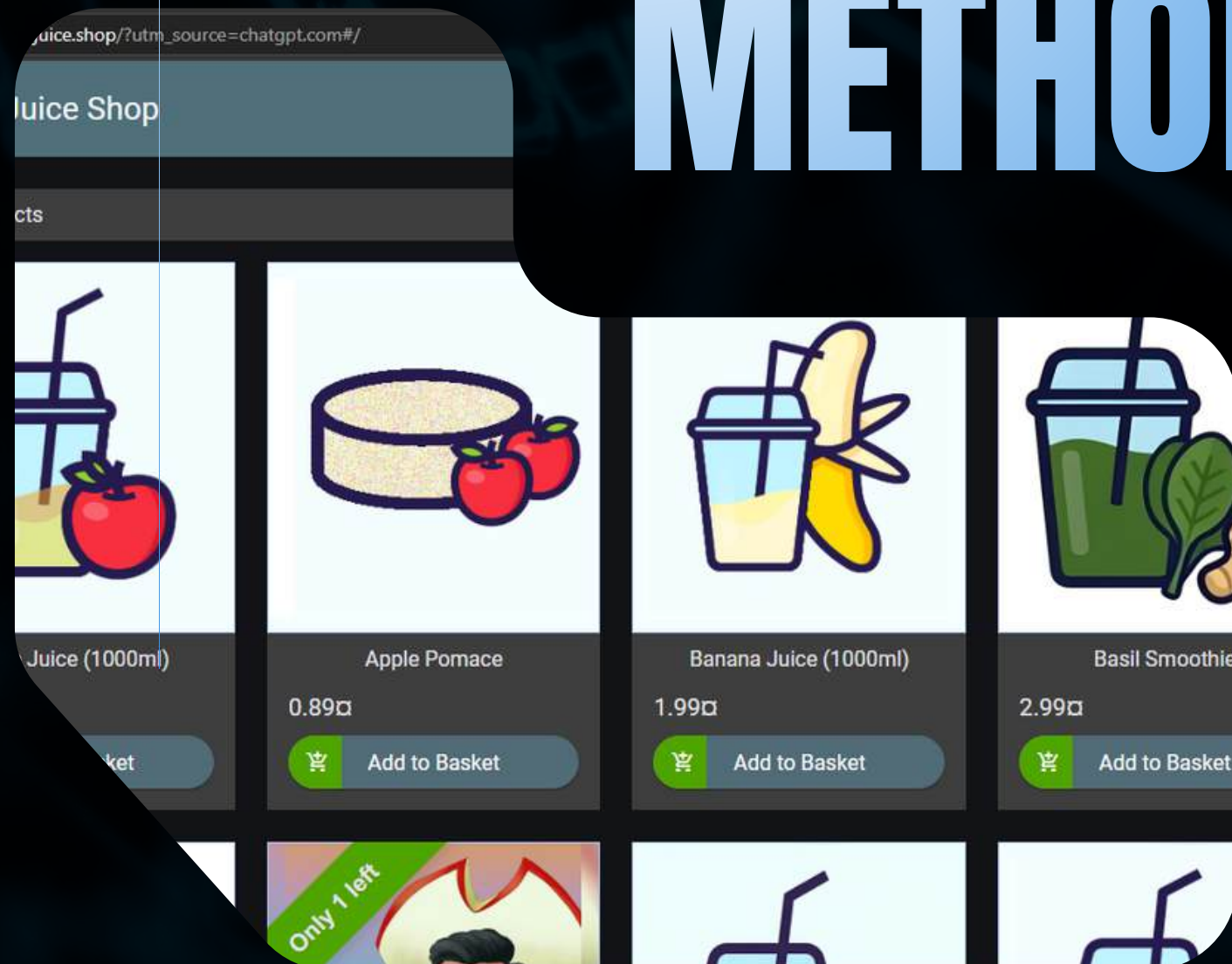
Why This Vulnerability Assessment Matters

1. Identifies critical security weaknesses before real attackers can exploit them
2. Protects web applications from data breaches, unauthorized access, and cyber threats
3. Helps organizations maintain confidentiality, integrity, and availability of sensitive data
4. Provides practical cybersecurity experience essential for real-world security roles

Key Impact:

Regular vulnerability assessments like this one on OWASP Juice Shop significantly reduce the risk of costly security incidents and strengthen overall security posture.

ASSESSMENT METHODOLOGY



Content:

The assessment followed these phases:

- Website reconnaissance and exploration
- Network and service discovery using Nmap
- Passive vulnerability scanning using OWASP ZAP
- Collection of security alerts and evidence
- Risk analysis and classification
- Documentation of findings and recommendations

Approach: Passive and non-destructive testing.

NMAP SCAN RESULTS

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\siyab>nmap demo.owasp-juice.shop
Starting Nmap 7.99 ( https://nmap.org ) at :
Nmap scan report for demo.owasp-juice.shop (
Host is up (0.19s latency).
Other addresses for demo.owasp-juice.shop (r
rDNS record for 81.169.145.156: w9c.rzone.de
Not shown: 990 filtered tcp ports (no-respons
PORT      STATE SERVICE
80/tcp    open  http
113/tcp   closed ident
443/tcp   open  https
1503/tcp  closed imtc-mcs
1720/tcp  closed h323q931
1935/tcp  closed rtmp
5060/tcp  open  sip
5222/tcp  closed xmpp-client
8010/tcp  open  xmpp
8042/tcp  closed fs-agent

Nmap done: 1 IP address (1 host up) scanned in 35.26 seconds
C:\Users\siyab>
```

Nmap Scan Findings

Content:

- HTTP service detected
- HTTPS service detected
- Active web server identified
- Publicly accessible services discovered

Importance: Network scanning helps understand the attack surface of the web application.

COMMON CYBER THREATS

Missing Security Headers

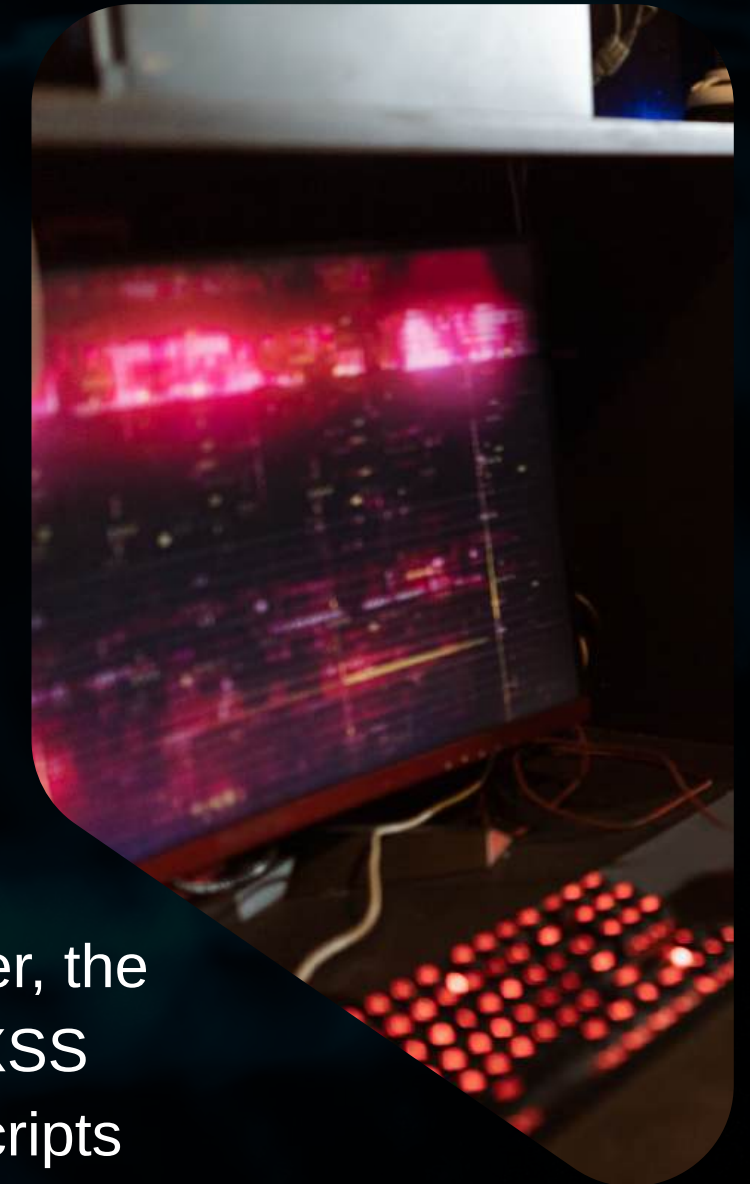
Absence of critical HTTP headers such as Content Security Policy (CSP) and X-Frame-Options, which were detected during the assessment.

Information Disclosure

The server is leaking technical details (server version, technologies used), making it easier for attackers to identify vulnerabilities and plan targeted attacks.

Cross-Site Scripting (XSS) Risk

Due to missing CSP header, the application is exposed to XSS attacks where malicious scripts can be injected and executed in users' browsers.

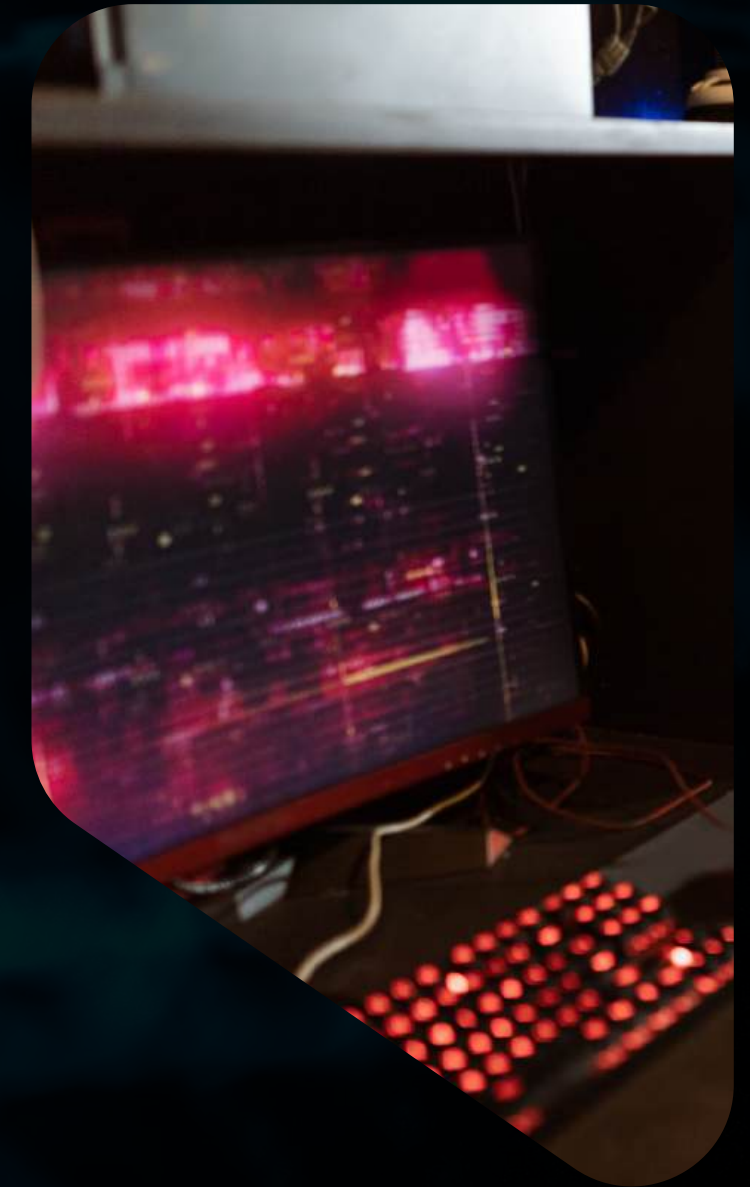


OWASP ZAP ANALYSIS

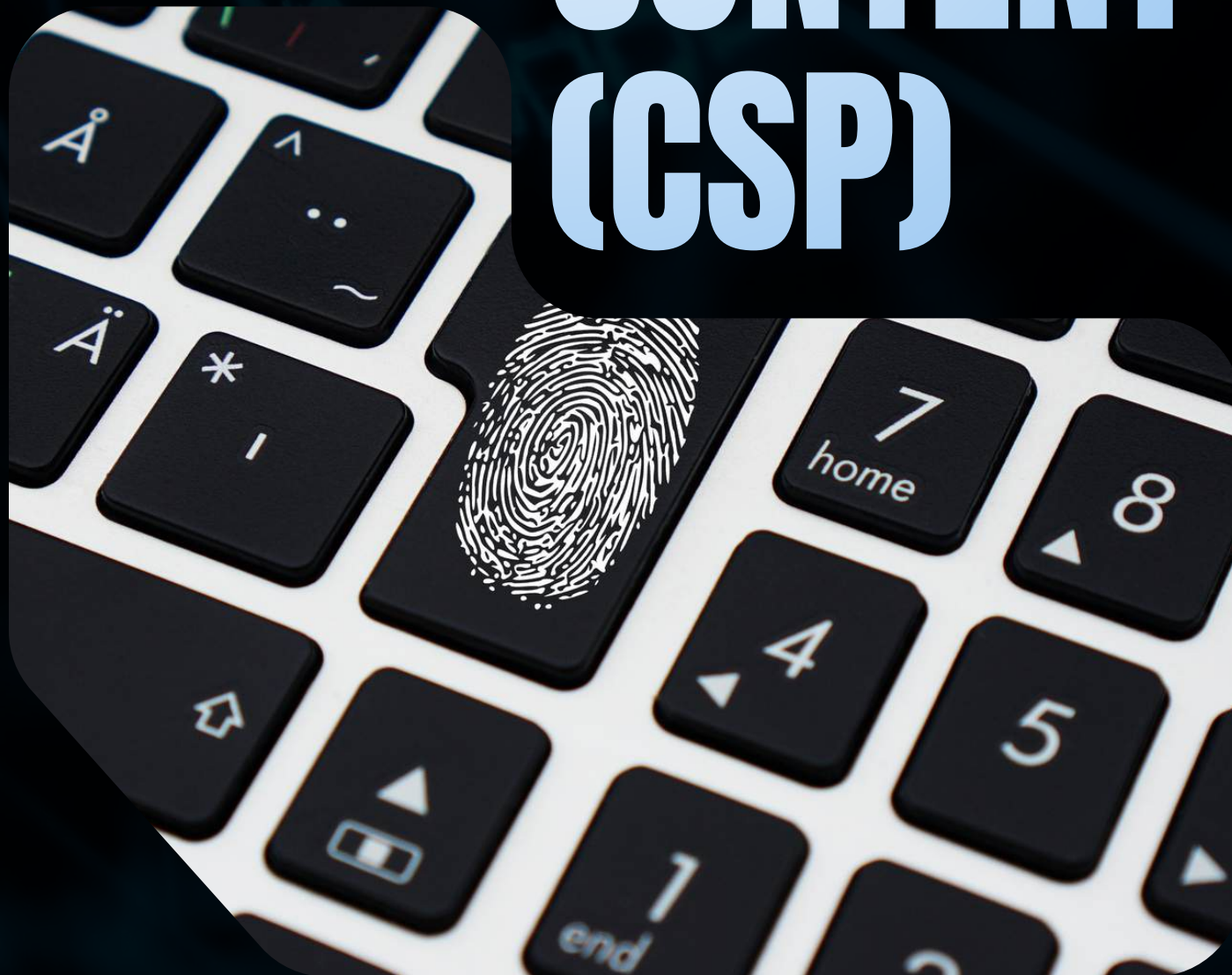
OWASP ZAP Vulnerability Analysis

Content:

- Performed passive vulnerability scanning on the target website
- Generated multiple security alerts including missing security headers and information disclosure
- Purpose: Identify vulnerabilities without actively exploiting the system



VULNERABILITY 1: MISSING CONTENT SECURITY POLICY (CSP)



Content:

- Risk Level: Medium
- Description: The application did not implement a proper Content Security Policy header.
- Impact: Increases risk of Cross-Site Scripting (XSS) attacks through malicious script injection.
- Recommendation: Implement a properly configured CSP header to restrict unauthorized scripts.

ADDITIONAL SECURITY FINDINGS

Content:

- Server information disclosure
- Missing security-related HTTP headers
- Weak client-side protections
- Potential exposure of technical information

Impact: These findings, though mostly low risk, contribute to a larger attack surface if left unaddressed.





RECOMMENDATIONS

Security Recommendations

Content:

- Implement secure HTTP security headers
 - Perform regular vulnerability assessments
 - Apply secure coding practices
-

- Keep systems and dependencies updated
 - Improve monitoring and logging
 - Conduct continuous security awareness and testing
-

Long-Term Benefit: Stronger security posture and reduced cyber risk.

CONCLUSION

Content:

The vulnerability assessment successfully identified key security weaknesses in the OWASP Juice Shop application using Nmap and OWASP ZAP.

This project provided valuable practical experience in cybersecurity testing, risk analysis, and professional reporting.

Thank You

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THANK YOU

